

EA Prerequisite: Fundamental Commissioning and Verification

This prerequisite applies to

- ▶ BD+C: New Construction
- ▶ BD+C: Core & Shell
- ▶ BD+C: Schools
- ▶ BD+C: Retail
- ▶ BD+C: Data Centers
- ▶ BD+C: Warehouses & Distribution Centers
- ▶ BD+C: Hospitality
- ▶ BD+C: Healthcare

INTENT

To support the design, construction, and eventual operation of a project that meets the owner's project requirements for energy, water, indoor environmental quality, and durability.

REQUIREMENTS

NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE

Commissioning Process Scope

Complete the following commissioning (Cx) process activities for mechanical, electrical, plumbing, and renewable energy systems and assemblies, in accordance with ASHRAE Guideline 0-2013 and ASHRAE Guideline 1.1-2007 for HVAC&R Systems, as they relate to energy, water, indoor environmental quality, and durability.

- ▶ Develop the OPR.
- ▶ Develop a BOD.

The commissioning authority (CxA) must do the following:

- ▶ Review the OPR, BOD, and project design.
- ▶ Develop and implement a Cx plan.
- ▶ Confirm incorporation of Cx requirements into the construction documents.
- ▶ Develop construction checklists.
- ▶ Develop a system test procedure.
- ▶ Verify system test execution.
- ▶ Maintain an issues and benefits log throughout the Cx process.
- ▶ Prepare a final Cx process report.
- ▶ Document all findings and recommendations and report directly to the owner throughout the process.

Requirements for exterior enclosures are limited to inclusion in the owner's project requirements (OPR) and basis of design (BOD), as well as the review of the OPR, BOD and project design. ASTM E2947-16: Standard Guide for Building Enclosure Commissioning provides additional guidance.

The review of the exterior enclosure design may be performed by a qualified independent member of the design or construction team (or an employee of that firm) who is not directly responsible for design of the building enclosure for the project.

Commissioning Authority Qualifications

By the end of the design development phase, engage a commissioning authority with the following qualifications.

- ▶ The CxA must have documented commissioning process experience on at least two building projects with a similar scope of work. The experience must extend from early design phase through at least 10 months of occupancy;
- ▶ The CxA may be a qualified employee of the owner, an independent consultant, an employee of the design or construction firm who is not part of the project's design or construction team, or a disinterested subcontractor of the design or construction team.
 - For projects smaller than 20,000 square feet (1 860 square meters), the CxA may be a qualified member of the design or construction team.

In all cases, the CxA must report his or her findings directly to the owner.

Current Facilities Requirements and Operations and Maintenance Plan

Prepare and maintain a current facilities requirements and operations and maintenance plan that contains the information necessary to operate the building efficiently. The plan must include the following:

- ▶ a sequence of operations for the building;
- ▶ the building occupancy schedule;
- ▶ equipment run-time schedules;
- ▶ setpoints for all HVAC equipment;
- ▶ set lighting levels throughout the building;
- ▶ minimum outside air requirements;
- ▶ any changes in schedules or setpoints for different seasons, days of the week, and times of day;
- ▶ a systems narrative describing the mechanical and electrical systems and equipment;
- ▶ a preventive maintenance plan for building equipment described in the systems narrative; and
- ▶ a commissioning program that includes periodic commissioning requirements, ongoing commissioning tasks, and continuous tasks for critical facilities.

Data Centers only

For small projects with computer room peak cooling loads less than 2,000,000 Btu/h (600 kW) or a total computer room peak cooling load less than 600,000 Btu/h (175 kW), the CxA may be a qualified employee of the design or construction team.

GUIDANCE

Refer to the LEED v4 reference guide, with the following additions and modifications:

Behind the Intent

Beta Update

Updated referenced standards.

Further Explanation

Commissioning Authority Qualifications

Delete the phrase "If the project is pursuing fundamental commissioning only" in the second paragraph.

In Table 1. Who can be the CxA:

- ▶ Revise the heading "be CxA for..." to "be CxA" in the right column
- ▶ Delete the subheading "fundamental Cx" in the right column
- ▶ Delete subheading "enhanced Cx" and all rows beneath it in the far right column of the table

Related Credit Tips

Refer to the LEED v4 reference guide, with the following modification:

- ▶ EA credit Renewable Energy. Renewable energy systems installed on-site must be commissioned under this prerequisite.

Referenced Standards:

- ▶ ASHRAE Guideline 0-2013, The Commissioning Process
- ▶ ASHRAE Guideline 1.1-2007, HVAC&R Technical Requirements for the Commissioning Process
- ▶ ASTM E2947 - 16: Standard Guide for Building Enclosure Commissioning

Connection to Ongoing Building Performance

- ▶ LEED O+M EA credit Energy Performance: Testing building systems after installation is fundamental to ensuring that systems function as designed. The development of a commissioning plan ensures that the building owner and facility managers have the information necessary to operate the building efficiently.